

RESEARCH ASSIGNMENT 2

THE LANGUAGE OF BUSINESS ANALYTICS

1) FOUNDATIONAL CONCEPTS OF ANALYSIS

1) Business Analytics, according to my understanding, is a practice that businesses would use to understand what is happening in their business, and how they can use this information to better or informed decisions in future.

A real world example where BA would apply is, a bakery sets a target of 5000 loaves over a period of a month, however the period ends without them having reached the target. The business can use business analytics to analyse their historical sales data to check the patterns or trend of sales over the set period, to understand why target was not reached.

IBM (2021) says that business analytics involves the use of data, by companies, which has been created by their operations to solve problems, monitor fundamentals and identify opportunities of growth to ultimately serve their customers better.

2) An article by Investopedia (2017) describe Data Analytics is a process of looking at raw data, using techniques that have been automated into mechanical processes that turn raw data into information that is useful. My understanding of this concept is that it allows the analysis of historical data that has been collected and turns into patterns, trends or reports, that can be understood by humans. I believe it has the same objectives as Business Analytics, however Data Analytics is not limited to just businesses.

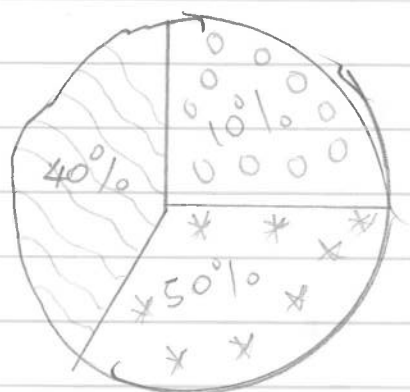
A real-world example of data analytics in action would be, a newly appointed detective wants a report of all cases related to gender-based violence, that have been reported in the area over the last 6 months. They can use the Crime Administration System database to track the cases, find out their nature and whether or not they have been solved. This analysis can help said detective to identify a course of action towards unresolved cases, and find way to reduce these crimes going forward.

03) My understanding of Business Intelligence (BI), is that the actual tools that are used to collect, manage and organize during a business analysis. According to IBM, Business Intelligence contains a set of technological tools that turn raw data into meaningful insights that may inform business strategies and operations.

An example where BI would be useful would be, say a company is competing with 2 others, and they all offer the same product or service. Company A wants determine the percentage of the market share, compared to Company B and Company C.

Company A can use Business Intelligence to access data from their own operations, third-party and external data to calculate their share of the industry

E.g - Pie charts



- ▣ Company A
- ▣ Company B
- ▣ Company C

04) The Harvard Business School (2019) defines Data-driven Decision-making as the process of using data as a way to inform decision-making processes and validating a course of action before committing to it.

My understanding of this concept is that DDDM is a practice that uses data as the primary influence on the decision-making process that a company follows, and that in order for it to work, the business needs to have a set of business objectives or goals.

Say a department in a business wants to improve its organizational health, but is not sure which key areas need improvement.

The department can conduct a survey, where employees give an input on the current state and what they think should be improved, in turn the business can use the data to actions that improve organizational health.

05) ~~Big data refers to the an~~

Big Data refers to a collection of all kinds of data that has been collected over time, and continues to grow fast. According to an article by Google Cloud, this is extremely large data and contains all types, whether structured, unstructured or semi-structured. Within the concept of Big Data are 3 characteristics known as the 3 V's of Big Data, with the first being Volume, which simply just how large the data is and from how many sources it has been collected. The second is Velocity and refers to the speed at which the data is being collected and sighting the importance of it being analyzed at the same speed. Lastly is variety which highlights the diversity of the data, inclusive of e.g spreadsheets, folders and images.

Because like most concepts and objectives in Data Analytics, the aim is to improve on decision-making and efficiency, and in the real world a business can use Big Data to influence its decisions.

However the key difference is that Big Data gives access to all sorts of data and in large amounts, meaning a business can use many types of data from all angles to make decisions.

6) At the forefront of Descriptive Analytics is the desire to understand what happened, and Investopedia says that descriptive analytics is the process of using historical data to understand the changes that have occurred in a business, most times using information from other companies for comparison.

In the case that a business is analysing its sales currently, compared to previous years, Descriptive Analytics can be used to answer the question 'What happened' and the data will answer whether sales improved or declined over the years.

7) My understanding of the Diagnostic Analytics concept is that its aim is to help understand why something has happened, and forces business or organizations to dig deep and come up with possible reasons or factors that may have contributed. Investopedia says that this concept tends to involve more data inputs and also a bit hypothesizing. Say hypothetically speaking the most recent Scorpion Kings concert was cancelled due to not enough ticket sales, the event organizers could have used Diagnostic Analysis to understand why ticket sales were low, could it have been that they were ^{not} affordable or could it be that they did ^{not} market the event properly.

08) Predictive Analytics is the third of the four types of data analytics and I believe it refers to the use of historical data to predict what could happen in future. This means there is no guarantee that something can happen, but should be anticipated, "Historical data is used to make assumptions about the future" (Harvard Business School Online, 2021). The most common example of the predictive analytics in the real world is the weather, which we can now access easily on our phones, and this helps us plan better because if I know it will rain tomorrow I can carry an umbrella, and if it might be sunny I can carry a sunhat and sunscreen.

09) Prescriptive Analytics, I believe leans more towards solutions, or possibilities, meaning that data is analysed with the aim of what is it that be done to meet objectives or goals. The Harvard Business School (2021) says that this concept is often called "the future of data analytics". In businesses, I believe this concept can be used to come up with possible solid actions, for example the organizers are now aware that their event was unsuccessful due to overpriced tickets or inadequate marketing. They can use Prescriptive Analytics to determine their target market and price tickets according to that, and invest more in their marketing, such as street posters and flyers if that was not a part previously.

10) According to IBM (n.d) Data Literacy is the ability to read, understand, use and communicate with data. I believe it simply means having the skill to read data, understand what it entails, the ways in which you can use it and how you can demonstrate your understanding of it. Seeing as we are currently in the 4IR, our real world is mostly digital and I believe data literacy stresses the importance of being able to interact with the digital world. It can be as simple as logging onto your laptop, being able to use systems such as banking apps or even ordering food online, to being able to code or use SQL.

02) Data, Measurement and Reporting

11) As previously established in class Structured Data is data that is usually easy to analyze because it follows a pre-defined format and is usually in a table that contains rows and columns.

A common use of structured data I use in the real world is a budget spreadsheet, which is a table that has been divided into two columns that specify expenses and cost, and rows that contain items and item cost.

12) Unlike structured data, Unstructured Data is free-flowing and is not contained by tables or tags, meaning it is not easy to analyse. ~~I believe that unstructured data in the real world for example conducting an analysis on how.~~

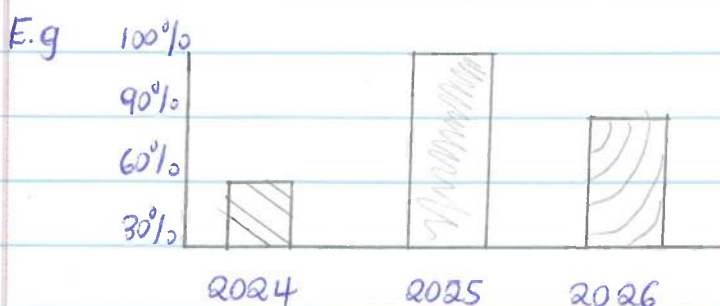
For example, say the Scorpion Kings event was a success and the organizers want to see the negatives and positives, and because most things are posted on social media. The organizers can check for social media posts, images and videos that mention the concert and filter for both positive experiences, check comments for any

criticism. This can help them understand what attendants enjoyed the most and which areas can they improve on.

13) As stated in the lesson on datawarehouses, they are a central data store, that houses all the data retrieved from all the multiple databases. I believe these serve as a larger place where all the clean data, making it ready to be analyzed by a data analytics. In the real world most companies have multiple branches, and I believe datawarehouse make it easy for business to house all the important data from several databases and store it in one place.

14) According to my understanding Data Mining is a process where historical data is analyzed for any valuable information or patterns that may have been missed. AWS says that data mining is crucial to businesses, and I believe this can apply in cases where companies are attempting to find new streams of revenue or regain customers, using data they have on hand.

15) IBM describes Data Visualization as the representation of data through the use of common graphic such as charts, plots and infographics amongst others. I believe this means turning the data into visuals that are easier to understand, read and follow, making different perspectives become useful to find a common objective. I believe companies can use data visualization to demonstrate, for instance their sales over a period of months or years.



16) Tableau describe a dashboard as a way of displaying various types of visual data all in one place, with the intention of conveying different but related data in a way that is easy to digest. I believe a dashboard is a an example of data visualization brought to life, and can be used by businesses to show data that shows performance in real time.

17) According to my know a metric is a system that is used to quantify things, such as cm, kilometres or litres; which is the system we use in South Africa. However in the context of data analytics, IBM describes metrics as something that allows you to consolidate measurements from various data analysis steps into single meaningful measurement. I believe this simply means turning data into percentages amongst other metrics, and companies use this system commonly by, for example, quantifying their sales over a period into one percentage.

18) Investopedia defines ^{KPI's} measurement used to assess a company's long-term performance. I believe that KPI are shown in the form of metrics but they apply specifically to businesses, as a way to indicate that the business is achieve its goals or objectives.

19) According to my knowledge a benchmark is a level or threshold to which others can be compared to, however in data analytics it is the process of measuring the performance of an organization's performance against standards in the industry or against competitors. I believe benchmarks are important to real-world ~~visualize~~ business as it helps them keep track of their performance, identify gaps and help them improve.

20) Segmentation as a concept refers to the dividing broad information into small parts, paying attention to similarities and characteristics. Investopedia says that market segmentation can be split into 4 types, namely demographic, geographic, behavioral or psychographic. I believe segmentation can be useful to businesses as it can help them identify their target market, and help them focus on the needs of those people specifically.

23) Customer & Sales KPIs

21) An article by Investopedia says that a conversion rate is the ratio between two currencies used in foreign exchange markets. I believe ~~this~~ this concept would apply in cases where you buy an item from another country, or you visit a foreign country and have to exchange your country's currency and turn it into another using the conversion rate.

22) In the digital marketing context, a conversion rate refers to the calculation in percentages of users who completed a specific action out of the total number of visitors to your website or marketing platform (Covrera, n.d). For example, when you visit a website and it asks you to sign up for a newsletter to get weekly updates, I believe a conversion rate can be used to determine if, of all the 100 visitors to the page, what percentage chose to sign up.

22) My understanding of the Customer Acquisition Cost concept is that it is the calculation of how much it would cost to acquire a customer in business. An article by Corporate Finance Institute (2019) says that CAC is calculated by sales, marketing expenses, divided by the number of new customers acquired over the measurement period.

For example a company launches a promo campaign that will span over a month, and the strategy includes hiring promoters, buying new merchandise, flyers and a billboard advert. Upon its completion the company can calculate all money spents on the campaign vs the customers the gained in the month to determine the cost of acquiring each customer.

23) Customer Lifetime Value, according to IBM is a calculation of the total profit from customer to business of the entirety of their relationship. I believe this concept refers to the amount of money a business has gained from a customer since they began using their product or service. It is also important to note that CLV can be measured using historic data, but can also be used in predictive form, meaning it looks how much a customer has spent and how much a customer will spend.

24) Investopedia says the the churn rate, is a calculation, in percentages of customers who discontinue their relationships over a period of time. I believe businesses can use this concept to determine their ability to keep customers/employees, and what the reasons could be. For example, a bakery can calculate their churn rate and see if customers are choosing to stop buying because of poor service or other reasons.

25) Customer Retention Rate happens to be the opposite of churn, and according to IBM, it refers to an organization's ability to keep customers over a period of time. I believe CRR can help businesses realize that they are doing something right, in terms of providing services.

26) In an article by Hubspot, Net Promoter Score is described as a customer loyalty metric that measures how likely customers are to recommend your business to others. I believe this helps business to determine how satisfactory is the service they provide and if not how can it be improved. I believe common examples of NPS in the real-world are surveys where we are asked what we would rate the service of a scale of 1-10, and if its low you are asked reasons why and how you would have liked it.

27) "A metric that tracks the average dollar amount spent on every order placed on a website or application" (CFI, 2019). This is a definition by CFI on the Average Order Value concept, and I believe it simply means the amount on average that a customer spends on orders, when ordering on apps or websites. CFI also says that this concept helps provide insights regarding customer behaviour. I believe this helps companies determine the revenue gained from each small or big transaction a customer makes

28) IBM describes a Sales Funnel as a visual roadmap of a customer's journey from initial awareness to final purchase, and that it is split into 6 stages namely Awareness, Interest, Decision, Intent, Evaluation and action, and lastly Loyalty and Retention. I think that a sales funnel just shows the stage a customer goes through before deciding to buying a product, and can includes research on the product, comparing it to similar products and even checking reviews before finally purchasing.

04) Marketing & Campaign Analytics

29) According to my knowledge, a marketing campaign is a planned set of efforts, usually over a set of time aimed at advertising a product or service. Marketing campaigns are usually have a clear objective and are targeted at a specific group of people.

Businesses can use marketing campaigns as a way to improve sales, gain customers or to even make customers aware of a product or service.

30) I think that the Post-Campaign Analysis refers to an analysis conducted after a campaign has ended, and measures what worked. Because a campaign requires a clear set of goals or objectives, a post-campaign analysis can help determine whether the goals were achieved, e.g. - did sales improve? did the company gain new customers, or was the specific target reached.

31) I think that Incremental Revenue refers to any extra profit or gains a business gets from running a campaign. In the case that a company runs a campaign with the aim of gaining a 100 new customers or R100 000 profit, and at the end of a campaign they have 120 new customers and R120 000. Any gains beyond their set target, I believe can be referred to an Incremental Revenue

32) Investopedia defines Return on Investment as a performance measure that is used to evaluate an investments efficiency of profitability. I would define ROI as the profit gained after investing in something, and in marketing ROI, I believed can be used to calculate or determine profit or gains achieved after spending money on a campaign or product.

32) Hubspot defines Return on Ad Spend as a measure of how much revenue is generated for every dollar spent on advertising campaigns. My understand of this concept is that it calculates revenue gained by a company compared to the amount spent on an ad. I believe this helps companies determine whether the money spent on advertising is beneficial.

34) Click-Through-Rate, according to Investopedia is a way to determine, through percentage, how effective an online ad engages viewers by calculating how many clicked on it after viewing. I believe CTR would count as a post-campaign analysis because say advert introducing a new product with the aim of reaching 1000 website visits, CTR can be used to determine whether campaign target was reached or what percentage ~~was reached~~ moved on to visit the website after watching the campaign ad.

35) I think that the Bounce Rate refers to the quantifying, in percentage, people that visit websites but leave quickly. I believe that is can also count as post campaign analysis, because if a company post a 30 second campaign ad, and in order it to count as a page-visit, the viewer would need to stay for atleast 15 seconds but leaves only after 10 seconds. The BR can help determine how effective was the ad campaign in holding the customer's attention and whether or not it worked.

36)

According to Sales Force, A/B Testing is method of comparing two versions of a marketing asset to see which one performs better.

I think that A/B testing means using two marketing efforts to see which one are users most receptive to. An article by Oracle says that it can also called split testing, meaning two content version are compared to find the better one.

Seeing as a campaign would require a set goal/objective, A/B testing can be used where different campaigns are put out at the same time, and their performance can be used to which one met the set goal/objective the most.

37) I think that Lead Generation is a process of attracting potential customers into buying a product or service. In sales lead generation is carried out through calls to customers, trade visits and promotions amongst others. I believe lead generation is a way for business to gain customers who may be interested but need convincing, but it can also help give insights on why said potential customers are not actual customers.

38) An article by Hubspot describes Marketing Attribution as a reporting strategy that allows marketing and sales teams to see the impact that marketing campaigns had on sales. I think that it is a way that analyses all goals/objectives of a marketing campaign and which efforts worked in achieving those set goals. If the marketing campaign of a company included TV ads, a billboard and several tik tok videos, marketing attribution helps determine which of these produced the most results.